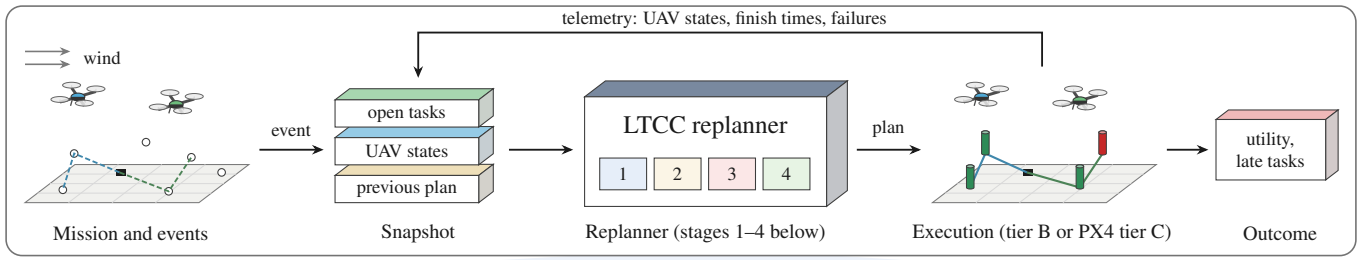
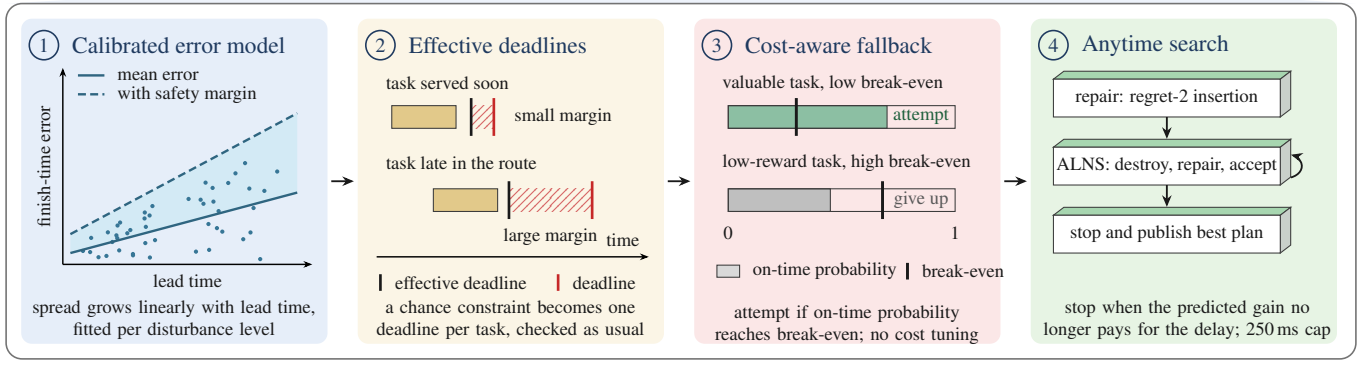

telemetry: UAV states, finish times, failures
wind
event
open tasks
UAV states
previous plan
LTCC replanner
1
2
3
4
plan
utility,
late tasks
Mission and events
Snapshot
Replanner (stages 1–4 below)
Execution (tier B or PX4 tier C)
Outcome


1 Calibrated error model
mean error
with safety margin
finish-time error
lead time
spread grows linearly with lead time,
fitted per disturbance level
2 Effective deadlines
task served soon
small margin
task late in the route
large margin
time
effective deadline
deadline
a chance constraint becomes one
deadline per task, checked as usual
3 Cost-aware fallback
valuable task, low break-even
attempt
low-reward task, high break-even
give up
0
1
on-time probability
break-even
attempt if on-time probability
reaches break-even; no cost tuning
4 Anytime search
repair: regret-2 insertion
ALNS: destroy, repair, accept
stop and publish best plan
stop when the predicted gain no
longer pays for the delay; 250 ms cap